Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

Instructions

This quiz will use the Coursera Lab Jupyter notebook "Lab to simulate the spring-mass-damper system" from this lesson.

In another browser tab or window, open the lab. You can run the lab as-is to learn how to implement a simulation of the spring-mass-damper system. However, you will modify several variables in the code before using the output from the code to answer the practice-quiz questions for this lesson.

In particular,

- Modify the block mass to be 10.
- Modify the damper constant to be 1.
- Modify the spring constant to be 2.
- Modify the number of simulations to be performed to be 1.

Make sure that you do NOT modify the line of code that states `randn("state",0)`. This must remain unchanged for the simulation to produce the results that are expected by the quiz autograder.

1. What is the final true position of the mass? (i.e., at time 100, at the end of the simulation.) Enter your answer with two digits to the right of the decimal point. **1 / 1 point**

-0.16195

Correct
Yes. That is correct.

2. What is the final true velocity of the mass? Enter your answer with two digits to the right of the decimal point. **1 / 1 point**

-0.26861

Correct
Yes. That is correct.

3. What is the final measured position $z(t)$ of the mass? Enter your answer with two digits to the right of the decimal point. **1 / 1 point**

-0.14886

Correct
Yes. That is correct.